Name:Harish Panpaliya

Branh:CS_AI

Division: D Batch-3

Roll no:52

PRN no: 12412173

Assignment 2:

Perofrm the following operations using Python on the data sets.

Compute and Display summary statistics for each feature available in the dataset.

Illustrate the feature distributions using

histogram

Data Cleaning,

Data Integration,

Data Transformation

collab Link

https://colab.research.google.com/drive/1bp12pKYafVc6O3TG3TxJiz8eAclANzud?usp=sharing

```
import numpy as np
from numpy import nan
import pandas as pd
```

```
df=pd.read_csv('Data_cleaning.csv')
df
```

	Row	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	1	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	2	01/02/2024	jane smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	3	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
3	4	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
4	5	2024-01-05	NaN	22.0	Miami	Monitor	300.0	2	600
5	6	2024-01-06	ALICE WONG	29.0	san francisco	Laptop	1200.0	1	1200
6	7	2024-01-07	Charlie Day	150.0	Philly	Mouse	25.0	10	250
7	8	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	9	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200

Data Cleaning

```
df = df.drop(columns=['Row'])
df
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	jane smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
3	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
4	2024-01-05	NaN	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	ALICE WONG	29.0	san francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	150.0	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200
9	2024-01-10	Natasha R.	32.0	miami	Mouse	25.0	2	50
10	2024-01-11	jane smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-01-12	Peter Parker	21.0	NYC	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	NaN	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	105.0	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington DC	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	106.0	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df = df.drop_duplicates()
df
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	jane smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
4	2024-01-05	NaN	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	ALICE WONG	29.0	san francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	150.0	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	?	50.0	3	150
10	2024-01-11	jane smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-13-01	Peter Parker	21.0	NYC	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	NaN	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	105.0	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington DC	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	106.0	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['Product'] = df['Product'].replace('?', np.nan)
df
```

```
/tmp/ipython-input-4083206129.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['Product'] = df['Product'].replace('?', np.nan)
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	jane smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	-5.0	Chicago	Keyboard	75.0	1	75
4	2024-01-05	NaN	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	ALICE WONG	29.0	san francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	150.0	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	jane smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-13-01	Peter Parker	21.0	NYC	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	NaN	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	105.0	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington DC	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	106.0	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df.loc[(df['Age'] <= 0) | (df['Age'] > 100), 'Age'] = np.nan
df
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	jane smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	NaN	Chicago	Keyboard	75.0	1	75
4	2024-01-05	NaN	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	ALICE WONG	29.0	san francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	NaN	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	jane smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-13-01	Peter Parker	21.0	NYC	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	NaN	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	NaN	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington DC	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	NaN	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['Customer_Name'] = df['Customer_Name'].str.title().fillna('Unknown')
df
```

```
/tmp/ipython-input-4222889279.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['Customer_Name'] = df['Customer_Name'].str.title().fillna('Unknown')
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	Jane Smith	34.0	LOS ANGELES	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	NaN	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29.0	san francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	NaN	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	NewYork	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-13-01	Peter Parker	21.0	NYC	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	NaN	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington DC	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	NaN	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['City'] = df['City'].str.title().str.strip()
city_mapping = {'Newyork': 'New York', 'Nyc': 'New York'}
df['City'] = df['City'].replace(city_mapping)
df
```

```
/tmp/ipython-input-3891373643.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['City'] = df['City'].str.title().str.strip()
/tmp/ipython-input-3891373643.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['City'] = df['City'].replace(city_mapping)
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	01/02/2024	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	NaN	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29.0	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	NaN	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	New York	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
11	2024-13-01	Peter Parker	21.0	New York	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28.0	Seattle	Laptop	1200.0	1	1200
16	18-Jan-24	Steve Rogers	NaN	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	Miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington Dc	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	NaN	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
df
```

```
/tmp/ipython-input-681347346.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	NaT	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	NaN	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29.0	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	NaN	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	NaN	1	80
8	2024-01-09	Edward Norton	40.0	New York	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
11	NaT	Peter Parker	21.0	New York	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28.0	Seattle	Laptop	1200.0	1	1200
16	NaT	Steve Rogers	NaN	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	Miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington Dc	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	NaN	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['Price'] = df['Price'].fillna(df['Total_Spent'] / df['Units'])
df
```

```
/tmp/ipython-input-3757453947.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['Price'] = df['Price'].fillna(df['Total_Spent'] / df['Units'])
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28.0	New York	Laptop	1200.0	1	1200
1	NaT	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	NaN	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22.0	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29.0	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	NaN	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31.0	New York	Headset	80.0	1	80
8	2024-01-09	Edward Norton	40.0	New York	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35.0	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34.0	Los Angeles	Mouse	25.0	2	50
11	NaT	Peter Parker	21.0	New York	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45.0	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38.0	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30.0	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28.0	Seattle	Laptop	1200.0	1	1200
16	NaT	Steve Rogers	NaN	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32.0	Miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	NaN	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3.0	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35.0	Washington Dc	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	NaN	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40.0	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38.0	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60.0	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df['Age'] = df['Age'].fillna(df['Age'].median()).astype(int)
df
```

```
/tmp/ipython-input-951792292.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['Age'] = df['Age'].fillna(df['Age'].median()).astype(int)
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28	New York	Laptop	1200.0	1	1200
1	NaT	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	34	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	34	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31	New York	Headset	80.0	1	80
8	2024-01-09	Edward Norton	40	New York	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
11	NaT	Peter Parker	21	New York	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28	Seattle	Laptop	1200.0	1	1200
16	NaT	Steve Rogers	34	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32	Miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	34	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35	Washington Dc	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	34	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df.to_csv('Cleaned_Data.csv', index=False)
df
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28	New York	Laptop	1200.0	1	1200
1	NaT	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	34	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	34	Philly	Mouse	25.0	10	250
7	2024-01-08	Diana Prince	31	New York	Headset	80.0	1	80
8	2024-01-09	Edward Norton	40	New York	Laptop	1200.0	1	1200
9	2024-01-10	Frank Castle	35	New York	NaN	50.0	3	150
10	2024-01-11	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
11	NaT	Peter Parker	21	New York	Web Cam	100.0	1	100
12	2024-01-14	Tony Stark	45	Malibu	Iron Suit	1000000.0	1	1000000
13	2024-01-15	Bruce Wayne	38	Gotham	Batarang	50.0	5	250
14	2024-01-16	Clark Kent	30	Metropolis	Glasses	15.0	1	15
15	2024-01-17	Unknown	28	Seattle	Laptop	1200.0	1	1200
16	NaT	Steve Rogers	34	Brooklyn	Shield	500.0	1	500
17	2024-01-19	Natasha R.	32	Miami	Mouse	25.0	2	50
18	2024-01-20	Wanda M.	34	Sokovia	Headset	80.0	1	80
19	2024-01-21	Vision	3	London	Battery	200.0	1	200
20	2024-01-22	Sam Wilson	35	Washington Dc	Wings	5000.0	1	5000
21	2024-01-23	Bucky Barnes	34	Brooklyn	Arm	15000.0	1	15000
22	2024-01-24	Scott Lang	40	San Francisco	Ant-Man Suit	0.0	1	0
23	2024-01-25	Hope Van Dyne	38	San Francisco	Wasp Suit	8000.0	1	8000
24	2024-01-26	Nick Fury	60	Unknown	Eye Patch	10.0	1	10

Next steps: Generate code with df New interactive sheet

```
df.describe()
```

	Date	Age	Price	Units	Total_Spent
count	21	24.000000	24.000000	24.000000	24.000000
mean	2024-01-14 09:08:34.285714176	33.041667	43098.333333	1.791667	43135.833333
min	2024-01-01 00:00:00	3.000000	0.000000	1.000000	0.000000
25%	2024-01-08 00:00:00	29.750000	25.000000	1.000000	68.750000
50%	2024-01-15 00:00:00	34.000000	90.000000	1.000000	225.000000
75%	2024-01-21 00:00:00	35.750000	1200.000000	2.000000	1200.000000
max	2024-01-26 00:00:00	60.000000	1000000.000000	10.000000	1000000.000000
std	NaN	9.958519	203847.874245	1.977683	203839.641890

```
df.head()
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
0	2024-01-01	John Doe	28	New York	Laptop	1200.0	1	1200
1	NaT	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
2	2024-01-03	Bob Brown	34	Chicago	Keyboard	75.0	1	75
4	2024-01-05	Unknown	22	Miami	Monitor	300.0	2	600
5	2024-01-06	Alice Wong	29	San Francisco	Laptop	1200.0	1	1200

Next steps: Generate code with df New interactive sheet

```
numeric_df = df.select_dtypes(include=[np.number])
numeric_df
```

	Age	Price	Units	Total_Spent
0	28	1200.0	1	1200
1	34	25.0	2	50
2	34	75.0	1	75
4	22	300.0	2	600
5	29	1200.0	1	1200
6	34	25.0	10	250
7	31	80.0	1	80
8	40	1200.0	1	1200
9	35	50.0	3	150
10	34	25.0	2	50
11	21	100.0	1	100
12	45	1000000.0	1	1000000
13	38	50.0	5	250
14	30	15.0	1	15
15	28	1200.0	1	1200
16	34	500.0	1	500
17	32	25.0	2	50
18	34	80.0	1	80
19	3	200.0	1	200
20	35	5000.0	1	5000
21	34	15000.0	1	15000
22	40	0.0	1	0
23	38	8000.0	1	8000
24	60	10.0	1	10

Next steps: Generate code with numeric_df New interactive sheet

```
summary_stats = pd.DataFrame({
    'Minimum': numeric_df.min(),
    'Maximum': numeric_df.max(),
    'Mean': numeric_df.mean(),
    'Range': numeric_df.max() - numeric_df.min(),
    'Standard Deviation': numeric_df.std(),
    'Variance': numeric_df.var(),
    '25th Percentile': numeric_df.quantile(0.25),
    '50th Percentile (Median)': numeric_df.quantile(0.50),
    '75th Percentile': numeric_df.quantile(0.75)
})

summary_stats.round(2)
```

	Minimum	Maximum	Mean	Range	Standard Deviation	Variance	25th Percentile	50th Percentile (Median)	75th Percentile
Age	3.0	60.0	33.04	57.0	9.96	9.917000e+01	29.75	34.0	35.75
Price	0.0	1000000.0	43098.33	1000000.0	203847.87	4.155396e+10	25.00	90.0	1200.00
Units	1.0	10.0	1.79	9.0	1.98	3.910000e+00	1.00	1.0	2.00

```
df['Price'].median()
```

```
90.0
```

```
df['Price'].mode()
```

	Price
0	25.0
1	1200.0

dtype: float64

```
df['Price'].min()
```

```
0.0
```

```
df['Price'].max()
```

```
1000000.0
```

```
df.nunique()
```

	0
Date	21
Customer_Name	22
Age	14
City	15
Product	16
Price	16
Units	5
Total_Spent	17

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 24 entries, 0 to 24
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           21 non-null     datetime64[ns]
 1   Customer_Name  24 non-null     object
 2   Age            24 non-null     int64
 3   City           24 non-null     object
 4   Product        23 non-null     object
 5   Price          24 non-null     float64
 6   Units          24 non-null     int64
 7   Total_Spent    24 non-null     int64
dtypes: datetime64[ns](1), float64(1), int64(3), object(3)
memory usage: 1.7+ KB
```

```
df.shape
```

```
(24, 8)
```

User Defined Functions

```
def calc_mean(numbers):
  sum_num=0;
  count=0;
  for i in numbers:
    sum_num=sum_num+i
    count+=1

  return sum_num / count


def calc_min(numbers):
  minimum=numbers[0]
  for i in numbers:
    if i<minimum:
      minimum=i
  return minimum
```

```
def calc_max(numbers):
  maximum=numbers[0]
  for i in numbers:
    if i>maximum:
      maximum=i
  return maximum



def calc_median(numbers):
    sorted_nums = sorted(numbers)
    n = len(sorted_nums)
    if n % 2 == 0:
        return (sorted_nums[n//2 - 1] + sorted_nums[n//2]) / 2
    else:
        return sorted_nums[n//2]


def calc_mode(numbers):
    freq = {}
    for num in numbers:
        freq[num] = freq.get(num, 0) + 1
    max_freq = max(freq.values())
    modes = [k for k, v in freq.items() if v == max_freq]
    return modes


def calc_variance(numbers):
    mean = calc_mean(numbers)
    summation=0
    for x in numbers:
      summation=summation+(x-mean)**2
    return summation / (len(numbers) - 1)


def calc_std(numbers):
    return calc_variance(numbers)**0.5
```

Comparison builtin vs Userdefined function outputs

```
df['Price'].mean()
```

```
np.float64(43098.333333333336)
```

```
k=calc_mean(df['Price'])
k
```

```
43098.333333333336
```

```
df['Price'].min()
```

```
0.0
```

```
k=calc_min(df['Price'])
k
```

```
0.0
```

```
df['Price'].max()
```

```
1000000.0
```

```
k=calc_max(df['Price'])
k
```

```
1000000.0
```

```
df['Price'].mode()
```

	Price
0	25.0
1	1200.0

dtype: float64

```
calc_mode(df['Price'])
```

```
[1200.0, 25.0]
```

```
df['Price'].var()
```

```
41553955834.057976
```

```
calc_variance(df['Price'])
```

```
41553955834.057976
```

```
import pandas as pd
import numpy as np

data = {
    'Date_Time': ['2026-01-01 10:00', '2026-01-01 11:00', '2026-01-01 12:00', '2026-01-01 13:00', '2026-01-01 14:00'],
    'Product_ID': [' A01', 'B02 ', 'A01', 'C03', 'B02'],
    'Category': ['Electronics', 'Home', 'Electronics', 'Garden', 'Home'],
    'Price': [100, np.nan, 120, 5000, 110], # 5000 is an outlier
    'Units_Sold': [10, 15, np.nan, 2, 18],    # Needs interpolation
    'Description': ['new model', 'old model', 'NEW model', 'damaged', 'old model']
}

df_f = pd.DataFrame(data)
df_f
```

	Date_Time	Product_ID	Category	Price	Units_Sold	Description
0	2026-01-01 10:00	A01	Electronics	100.0	10.0	new model
1	2026-01-01 11:00	B02	Home	NaN	15.0	old model
2	2026-01-01 12:00	A01	Electronics	120.0	NaN	NEW model
3	2026-01-01 13:00	C03	Garden	5000.0	2.0	damaged
4	2026-01-01 14:00	B02	Home	110.0	18.0	old model

Next steps: Generate code with df_f New interactive sheet

```
df_f.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date_Time    5 non-null      object
 1   Product_ID   5 non-null      object
 2   Category     5 non-null      object
 3   Price        4 non-null      float64
 4   Units_Sold   4 non-null      float64
 5   Description  5 non-null      object
dtypes: float64(2), object(4)
memory usage: 372.0+ bytes
```

```
df_f.describe()
```

	Price	Units_Sold
count	4.000000	4.000000
mean	1332.500000	11.250000
std	2445.013633	6.994045
min	100.000000	2.000000
25%	107.500000	8.000000
50%	115.000000	12.500000
75%	1340.000000	15.750000
max	5000.000000	18.000000

```
df_f['Units_Sold'].interpolate()
```

	Units_Sold
0	10.0
1	15.0
2	8.5
3	2.0
4	18.0

dtype: float64

```
df_f['Product_ID'].str.strip()
df_f['Description'].str.lower()
```

	Description
0	new model
1	old model
2	new model
3	damaged
4	old model

dtype: object

```
df_f['Price'].clip(lower=0, upper=200)
```

	Price
0	100.0
1	NaN
2	120.0
3	200.0
4	110.0

dtype: float64

```
df_f.set_index('Date_Time',inplace=True)
```

```
df_f.rename(columns={'Date_Time': 'Timestamp'},inplace=True)
```

```
df_f
```

Date_Time	Product_ID	Category	Price	Units_Sold	Description
2026-01-01 10:00	A01	Electronics	100.0	10.0	new model

Next steps: Generate code with df_f New interactive sheet

```
df_sample=df.sample(10)
df_sample
```

	Date	Customer_Name	Age	City	Product	Price	Units	Total_Spent
2026-01-01 10:00		Joe		Garden	5000.0	2.0	damaged	
10	2024-01-11	Jane Smith	34	Los Angeles	Mouse	25.0	2	50
5	2024-01-06	Alice Wong	29	San Francisco	Laptop	1200.0	1	1200
6	2024-01-07	Charlie Day	34	Philly	Mouse	25.0	10	250
18	2024-01-20	Wanda M.	34	Sokovia	Headset	80.0	1	80
21	2024-01-23	Bucky Barnes	34	Brooklyn	Arm	15000.0	1	15000
14	2024-01-16	Clark Kent	30	Metropolis	Glasses	15.0	1	15
24	2024-01-26	Nick Fury	60	Unknown	Eye Patch	10.0	1	10
19	2024-01-21	Vision	3	London	Battery	200.0	1	200
23	2024-01-25	Hope Van Dyne	38	San Francisco	Wasp Suit	8000.0	1	8000
4	2024-01-05	Unknown	22	Miami	Monitor	300.0	2	600

Next steps: Generate code with df_sample New interactive sheet

```
df['Date'].dt.year
```

	Date
0	2024.0
1	NaN
2	2024.0
4	2024.0
5	2024.0
6	2024.0
7	2024.0